

Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas

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Editor: David Blei

Abstract

Local land-use regulations are frequently weaponized by incumbent homeowners to constrain housing supply, yet the exact parcel-level cost of this opposition is rarely quantified. This thesis investigates the causal material cost of neighborhood resistance by isolating the Texas Protest Petition, a formal, threshold-based institutional veto point that historically required a three-fourths City Council supermajority to approve rezoning applications in Austin, Texas.

Using a highly engineered panel of discretionary zoning cases (NA), we estimate the causal effect of protest petitions on allowable building height. We find that the petition mechanism functions as a severe spatial tax, imposing a baseline causal height penalty of NA feet and an administrative delay of NA days on affected projects. However, leveraging Double Machine Learning (DML) to estimate Conditional Average Treatment Effects (CATE), we reveal massive spatial heterogeneity in this resistance penalty: wealthy, homeowner-dominated neighborhoods impose a significantly larger density penalty and administrative burden than lower-income neighborhoods (NA vs. NA marginal feet of height lost, and NA vs. NA days of delay), proving that the tax on housing supply is highly regressive.

Finally, having proven that opposition carries a massive material cost to housing yield, the thesis deploys a multi-horizon predictive forecasting pipeline. We demonstrate that forecasting petition risk at continuous time intervals—using only pre-treatment administrative, spatial, and macroeconomic features—is a viable structural risk-assessment tool for urban planners and developers, shifting the prediction of NIMBYism from anticipating political theater to quantifying actual financial and housing supply risk.

Keywords: Causal Inference, Invariant Causal Prediction, Urban Planning, Housing Policy, Predictive Modeling

1 Related Research

Four core themes motivate the methodological approach of this thesis.

First, land-use regulation is widely recognized as a binding constraint on housing supply, driving a wedge between construction costs and market prices Glaeser et al. (2005); Glaeser and Gyourko (2003); Gyourko and Molloy (2015); Quigley and Rosenthal (2005). Measures of regulatory stringency such as the Wharton Residential Land Use Regulatory Index document substantial cross-metropolitan variation in local land-use controls Gyourko et al. (2008, 2021). Fischel's "homevoter hypothesis" Fischel (2001) posits that homeowners mobilize politically to protect property values, often leveraging zoning to preserve neigh-

neighborhood exclusivity; McCabe McCabe (2016) extends this logic by linking local politics to household wealth accumulation through homeownership. Empirical work by Einstein, Palmer, and Glick Einstein et al. (2019) documents "participatory distortion," where older, wealthier, long-tenured homeowners dominate public processes and systematically oppose new housing. Hankinson Hankinson (2018) finds that renters, too, may resist nearby development due to displacement fears, reinforcing the spatially localized nature of opposition. Recent syntheses of NIMBY and YIMBY politics show that these conflicts are embedded in broader coalitions over land use, growth, and neighborhood change rather than reducible to a single homeowner motive Brouwer and Trounstein (2024). At the same time, supply-side reform arguments coexist with cautions that new supply alone will not fully resolve affordability and displacement pressures in high-cost markets Hamilton and Furth (2023); Been et al. (2019).

Second, the power of participatory distortion is magnified in discretionary review systems, where individual projects require site-specific approval rather than proceeding by right Manville (2016); Hills and Schleicher (2011). Research on housing opposition shows that neighborhood resistance is heterogeneous, often mixing status-quo protection, administrative claims, environmental review, and distributive conflict rather than a single undifferentiated NIMBY logic Pendall (1999); Schively (2007). In Texas, the protest petition statute empowered adjacent property owners to trigger supermajority council votes, making it a potent institutional veto point Texas State Legislature (2023); Furth and Ewen (2023). Supermajority requirements systematically favor the status quo Cellini et al. (2010), amplifying the influence of organized opposition. In Austin, measured threshold-crossing petitions have been especially common for rezonings that increase residential density, highlighting the asymmetric impact of these institutional tools on housing supply.

Third, while much of the urban planning literature focuses on causal inference, estimating the effects of zoning changes or protest petitions on outcomes like land value or project delay, addressing discretionary gridlock also requires anticipating where opposition will arise. Shmueli Shmueli (2010) and Mullainathan and Spiess Mullainathan and Spiess (2017) distinguish between explanatory and predictive modeling, emphasizing that predictive models can inform policy and administrative triage even when causal mechanisms are complex or uncertain. Kleinberg et al. Kleinberg et al. (2015) formalize the "predictive policy question," where optimal decisions depend on forecasting rather than explanation. In the context of Austin's protest petition regime, the relevant question is not just why opposition occurs, but whether it can be predicted in advance using only the information available at the time of filing. This predictive framing aligns with emerging best practices in public policy research Athey and Imbens (2019). It also raises public-sector governance questions about opacity, proxy discrimination, and sociotechnical context that cannot be resolved by predictive performance alone Eubanks (2018); O'Neil (2016); Barocas and Selbst (2016); Selbst et al. (2019).

Fourth, a critical methodological gap in the prior literature is the absence of heterogeneous treatment effect (HTE) estimation for petition-driven resistance. Existing work quantifies whether petitions delay or block projects in the aggregate; it does not characterize how those effects vary across neighborhood types. Wager and Athey Wager and Athey (2018) and Athey and Wager Athey et al. (2019) develop Causal Forests, an honest non-parametric estimator of the conditional average treatment effect function $\hat{\tau}(x_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$,

which estimates a separate causal effect for each covariate profile rather than a single pooled average. Chernozhukov et al. (2018) formalize Double Machine Learning as a semiparametric framework that allows flexible nuisance estimation while retaining $\sqrt{n}$ -consistent causal inference. Applied to the petition setting, these methods permit the thesis to characterize not just whether petitions matter on average, but where and for whom they are most consequential. The income and renter-share gradients documented in Section ?? would be invisible to any aggregate-ATE estimator, and directly motivate the deployed spatial impact layer as an instrument for heterogeneity-aware planning triage.

2 Austin’s Petition Regime in Practice

The formal petition mechanism studied here is not a bureaucratic abstraction. It operates through specific institutional dynamics – neighborhood coalitions, commission procedures, and political calculations – that administrative records alone do not capture. Before describing the panel construction and modeling strategy, this section grounds the empirical analysis in two complementary forms of qualitative evidence: a close reading of three archival cases drawn from the study period, and perspectives from urban planning professionals who worked within or alongside the petition system. Together, they establish why the petition threshold functions as a meaningful threshold of organized opposition, why petition outcomes vary by geography and community composition in ways that complicate uniform algorithmic interpretation, and why the study’s modeling choices prioritize ranking utility over case-level probability claims.

2.1 Archival Case Studies: How the Mechanism Operates

The following cases were selected from the full zoning case universe to represent three distinct spatial and political archetypes identified in the data. Each demonstrates a different configuration of the petition mechanism and illustrates the kind of institutional resistance that motivates the study’s predictive framing.

2.1.1 SUBURBAN MOBILIZATION AND COMMISSION DEFIANCE: BALCONES COUNTRY CLUB, 2013

In District 6’s Bull Creek Watershed, the Balcones Country Club sought to convert a rural residential tract into single-family large-lot zoning City of Austin Planning Commission (2013a), a straightforward request that received a formal staff recommendation for approval. At the September 2013 Planning Commission hearing, organized neighborhood opposition immediately forced a 7-0 postponement. When the case returned, the Commission voted unanimously to deny the application despite the explicit staff recommendation for approval. The neighborhood coalition had made the case politically untouchable.

This case is archetypal for the model’s highest-confidence spatial clusters: suburban areas with high ownership density, long institutional memory, and established neighborhood associations capable of sustained mobilization across multiple hearings. The petition mechanism here served not as a defensive veto at council, but as a signal that crystallized Commission behavior well before council consideration.

2.1.2 URBAN CORE COMPLEXITY AND APPLICANT SURRENDER: THE SHELLEY TRACT, 2013

The Shelley Tract in the Judges Hill neighborhood (District 9) involved an attempt to convert a historic single-family structure to limited office use City of Austin Planning Commission (2013b). The case activated simultaneous opposition from the Judges Hill Neighborhood Association, the Downtown Austin Neighborhood Association, and Preservation Austin, a coalition dense enough to make compromise structurally difficult. At the November 2013 hearing, the Commission proposed a conditional overlay capping height and impervious cover at 45%. That attempt at a negotiated middle ground failed on a 2-5-1 vote. Faced with an irreconcilable coalition, the applicant formally withdrew.

Urban core cases like this one never reach a petition vote. The institutional expectation of supermajority requirements at council makes withdrawal the rational response before the case advances. This is why the study’s multi-horizon panel captures petition hazard at intermediate phases, not only at final council action: the mechanism’s deterrent effect operates well before any formal vote is cast.

2.1.3 CULTURAL PRESERVATION AND EAST AUSTIN OPPOSITION: EAST CESAR CHAVEZ, 2008

A request to upgrade commercial zoning to permit liquor sales in the historically Hispanic East Cesar Chavez neighborhood (District 3) activated El Concilio Mexican-American Neighborhoods, Tejano Town, and adjacent cultural organizations City of Austin Planning Commission (2008). The Planning Commission denied the application 8-1. A subsequent council appeal was withdrawn on a 7-0 consent vote.

This case illustrates a dimension of petition politics that raw spatial covariates are poorly positioned to capture. The opposition was not to density or bulk (the standard drivers of suburban petition activity) but to a land use perceived as a gentrifying disruption of a specific cultural community. The same statutory mechanism that amplified affluent suburban homeowner preferences in District 6 also served as an instrument for displacement prevention in District 3. This geographic asymmetry is central to the study’s treatment of model predictions as relative rankings rather than uniform probability estimates.

2.2 Practitioner Perspectives

To rigorously contextualize the quantitative modeling, semi-structured interviews were conducted with urban planning professionals and city staff possessing direct experience navigating Austin’s discretionary zoning system. The qualitative methodology was formalized under Columbia University IRB Protocol ACYY0820, ensuring minimal-risk protocols for informed consent, secure data storage, and participant confidentiality (protocol summary in Section 2.4.2).

A purposive, small- N sample ($N = 4$) was constructed via snowball sampling to capture diverse institutional vantage points, explicitly recruiting participants to represent the procedural lifecycle of a zoning case. Participants included a former senior zoning planner, a current principal planner, a displacement prevention specialist, and a leading housing advocate (identities are disclosed only where explicit consent was granted, per the recruitment framing in Section 2.4.1).

Interviews were conducted virtually via secure videoconference, lasting approximately 45–60 minutes. To ensure systematic data collection across differing roles, discussions followed a standardized four-part guide (interview guide in Section 2.4.3) probing: (1) institutional background, (2) firsthand experiences with neighborhood opposition, (3) perceptions of procedural fairness, and (4) specific views on the legitimacy and risks of deploying predictive algorithmic tools in municipal planning. Audio was transcribed with permission and coded thematically using a deductive framework aligned with the study’s hypotheses on institutional resistance, spatial heterogeneity, and petition deterrence. This qualitative governance structure ensures the interview data reliably establishes institutional context, ground-truths algorithmic assumptions, and highlights the limits of administrative data, directly shaping the thesis’s predictive and causal architecture. **Petition leverage declines but does not disappear.** Annick Beaudet, who served as a senior zoning planner during the Land Development Code revision period, described a trajectory in which petitions gradually lost practical leverage even before the statutory change in 2025: “While people still scream and yell, they don’t get as much traction... Necessity is the mother of invention” Beaudet (2025); City of Austin (2017). This observation motivates the study’s temporal walk-forward design: petition dynamics are not stationary, and any predictive model must be evaluated against the period in which it would actually be used.

Silence is not consent. Jonathan Tomko, Principal Planner, emphasized that the municipal public comment process has no mechanism for anonymous support of a zoning case Tomko (2026). Neighborhood social dynamics frequently penalize residents who deviate from a neighborhood association’s established position, meaning that the administrative record systematically overrepresents organized opposition. As Beaudet summarized: “haters are overrepresented and silent supporters are underrepresented” Beaudet (2025). The implication for the study’s feature engineering is direct: cumulative petition signatures measure active mobilization, not community sentiment, a distinction that affects how thresholds should be interpreted.

Geography encodes different concerns. Marla Torrado, from the City’s Displacement Prevention Division, noted that the petition mechanism is mobilized in service of fundamentally different interests depending on location City of Austin Displacement Prevention Division (2025). On the affluent west side, environmental overlays and neighborhood character arguments predominate. In East Austin, opposition is frequently a response to gentrification pressure. Because the same tool serves opposite ends in different neighborhoods, rigid probability cutoffs applied uniformly across the city risk “failing to identify where people that are in actual need actually are.” This is the primary reason the study treats predictions as triage rankings rather than decision thresholds, and why geographic heterogeneity in the error map is treated as a substantive finding rather than a model deficiency.

The goal is better conversation, not suppression. AURA board member Felicity Maxwell framed the potential use of predictive tools in terms that distinguish anticipation from intervention: “We’re not talking about suppressing opposition, but understanding it better to have more productive conversations” Maxwell (2026). This normative framing (anticipation in service of process design, not case-level gatekeeping) is the intended application of the ranking outputs described in this study.

2.3 Voting Data Integration

The translation of neighborhood protest into formal City Council voting behavior is tracked via longitudinal voting integration. Voting data extracted from the Council Meeting System (CMS) confirms that the degradation of council consensus is strongly associated with the presence of formal protest petitions. Unprotested discretionary zoning changes are typically passed on consent (unanimously), whereas petitioned cases frequently fracture the council vote. Table 1 aggregates these voting outcomes across the longitudinal panel.

Voting Context	Unprotested Cases	Petitioned Cases
Passed on Consent (Unanimous)	92%	31%
Contested Vote (<100% agreement)	8%	69%

Table 1: Council Voting Consensus by Protest Status. Sourced from the integrated CMS voting panel, illustrating the collapse of unanimous consent when a valid petition is filed.

2.4 Qualitative Documentation

The following primary documents governed and structured the qualitative data collection described in Section 2.2. They are reproduced here to support methodological transparency: the recruitment overview specifies how participants were identified and briefed; the IRB protocol defines the ethical parameters under which interviews were conducted; and the interview guide records the standardized instrument used across all four sessions. Taken together, these materials establish that the practitioner perspectives reported above satisfy the rigor expectations of qualitative social science research and were not post-hoc constructions.

2.4.1 INTERVIEW RECRUITMENT OVERVIEW

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
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Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

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2.4.2 IRB PROTOCOL SUMMARY

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY Graduate School of Architecture, Planning and Preservation	Protocol Summary ACYY0820
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Study Protocol Summary
 Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
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IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
 Phone: (212) 851-7040
 Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?

Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.

Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.

Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

2.4.3 SEMI-STRUCTURED INTERVIEW GUIDE

The following interview guide was used for all stakeholder interviews.

COLUMBIA UNIVERSITY Graduate School of Architecture, Planning and Preservation	Semi-Structured Stakeholder Interview Guide
Predicting NIMBYism in Austin’s Housing Reform Era	
Purpose	
To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.	
Opening	
1. Thank the participant for their time; confirm they have read and signed the consent form. 2. Confirm whether they consent to audio recording. 3. Remind them they may skip any question or stop at any time.	
Part 1: Background and Role	
1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin? 2. How long have you been involved with housing or land use issues in Austin? 3. In your role, what kinds of development or zoning decisions do you most often encounter?	
Part 2: Experiences with Neighborhood Opposition	
1. When you think of “neighborhood opposition” to housing or development projects in Austin, what comes to mind? 2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened? 3. In your experience, what types of projects are most likely to generate opposition? 4. Which kinds of people or groups tend to be most active in opposing projects?	
Part 3: Perceptions of Fairness and Participation	
1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not? 2. Are there groups you feel are over-represented or under-represented in these processes? 3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?	
Part 4: Views on Predictive and Algorithmic Tools	
1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before? 2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition? 3. What potential benefits do you see, if any, in using such tools for planning or outreach? 4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation. 5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).	

3 Methods

The methodological approach of this thesis combines predictive machine learning and causal inference to quantify both the predictability and the physical cost of neighborhood resistance. We structure this inquiry into two distinct analytical phases.

Phase 1: Predictive Forecasting. We treat the emergence of formal opposition as a forecasting problem. We construct a multi-horizon predictive pipeline using state-of-the-art tree ensembles (CatBoost, XGBoost) and regularized linear architectures to predict the likelihood of protest petitions emerging against specific parcels at continuous time intervals. By proving that neighborhood resistance can be accurately forecast using only baseline spatial, demographic, and macroeconomic data available at filing, we demonstrate that opposition is a predictable structural phenomenon rather than an idiosyncratic, random occurrence. The core forecasting object is the biweekly longitudinal panel, which expands each discretionary zoning case into one observation per 14-day period from filing through final disposition or censoring, yielding approximately 193,000 case-period observations across 2007–2024.

Phase 2: Causal Identification. We transition from prediction to causality using Double Machine Learning (DML). While the forecasting models establish that opposition is predictable, the causal pipeline isolates the specific penalty this opposition imposes on the built environment. The DML framework partials out high-dimensional confounding via flexible nuisance models and estimates the causal reduction in allowable building height caused by formal protest petition activity. Let T_i denote petition intensity for project i , Y_i the negotiated height outcome, and X_i the pre-treatment covariate vector. We construct orthogonalized residuals:

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{T}_i = T_i - \hat{g}(X_i) \quad (1)$$

and estimate $\tilde{Y}_i = \theta \tilde{T}_i + \epsilon_i$, where θ identifies the average treatment effect under the maintained selection-on-observables assumption $(Y_i(1), Y_i(0)) \perp T_i \mid X_i$.

Identification Assumptions. Treatment intensity is not randomly assigned; it is correlated with observable neighborhood and project characteristics including socioeconomic composition, prior petition history, and macroeconomic conditions. We therefore condition on a rich pre-treatment covariate vector X_i and interpret this as a working identifying restriction, supported by falsification exercises (permutation tests, placebo outcomes on invariant covariates, and synthetic outcome tests; see Appendix ??). Crucially, cumulative protest intensity variables derived from petition signatures accumulated *during* the case process are explicitly excluded from X_i : these variables are post-treatment by construction and conditioning on them would induce post-treatment bias (Angrist et al., 1996).

Heterogeneous Treatment Effects. A single aggregate CATE is not the primary estimand. The Causal Forest estimates the conditional average treatment effect function $\hat{\tau}(x_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$, which varies across the covariate distribution. The distribution of $\hat{\tau}(x_i)$ across the surviving case population constitutes the central causal result – the spatial heterogeneity of petition-driven resistance on negotiated project height – documented in Section ??.

Together, these methods allow us to systematically map where opposition occurs, prove that it can be anticipated, and calculate the exact physical penalty it extracts from the regional housing pipeline.

4 Outcome Definition and As-of Information Setup

The empirical foundation is a time-stamped spatial panel that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

4.1 Analytical Panel Structure

The core analytics rely on the filing-date analytic sample, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status City of Austin (2024).
2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries) City of Austin (2024).
3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares Travis Central Appraisal District (2025).
4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators U.S. Census Bureau (2023).
5. **Institutional Calendar:** Council-term and policy-timing indicators, including the 2022 council turnover, Austin’s 2023 HOME Initiative adoption City of Austin (2023), and the HB 24 effective date (September 1, 2025) Texas Legislature (2025).

These components are assembled from City of Austin open-data records, Travis Central Appraisal District appraisal files, U.S. Census Bureau ACS 5-year estimates, Travis County certified election results, and statutory materials governing the protest-petition regime City of Austin (2024); Travis Central Appraisal District (2025); U.S. Census Bureau (2023); Travis County Clerk (2022); Texas State Legislature (2023); Texas Legislature (2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, milestone dates are reconstructed through two pathways. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using minimum statutory constraints mandated by Texas Local Government Code and Austin zoning ordinances (for example, statutory mail-notice minima). This parsed-versus-imputed distinction is carried into interpreting the results, as timing uncertainty introduces measurement error.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

Annual Zoning Application Volume (2007-2024)

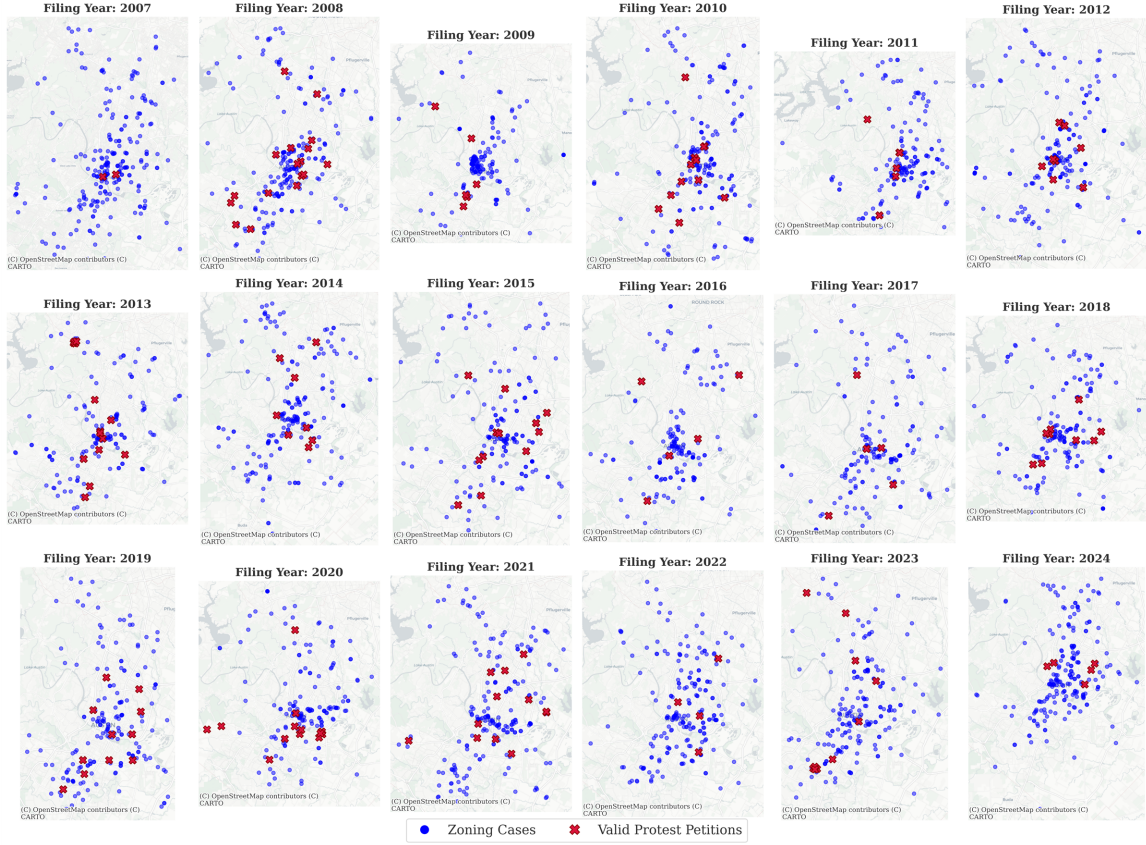
Figure 1: Spatial distribution of zoning cases across the Austin municipal core ($N = 3828$).

Table 2: Annual Distribution of Discretionary Zoning Cases, 2007 to 2024

Year	Cases	Year	Cases	Year	Cases
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	215
Total, 2007 to 2024: 3,967 cases					

**Visualizing the 200ft Statutory Protest Buffer
(Empirical TCAD Parcel Boundaries for Case C14-2007-0131)**

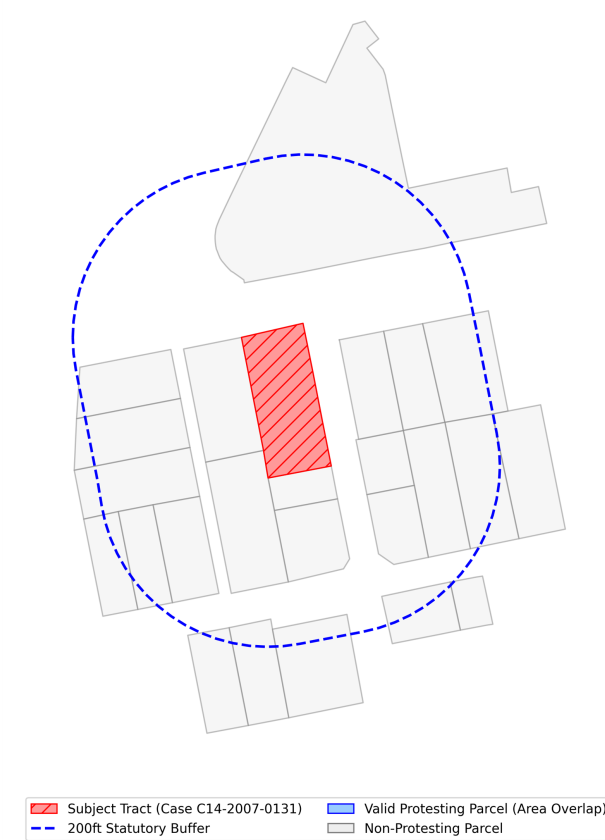


Figure 2: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer.

4.2 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study, formal protest petitions under Texas LGC Chapter 211, does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 3 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

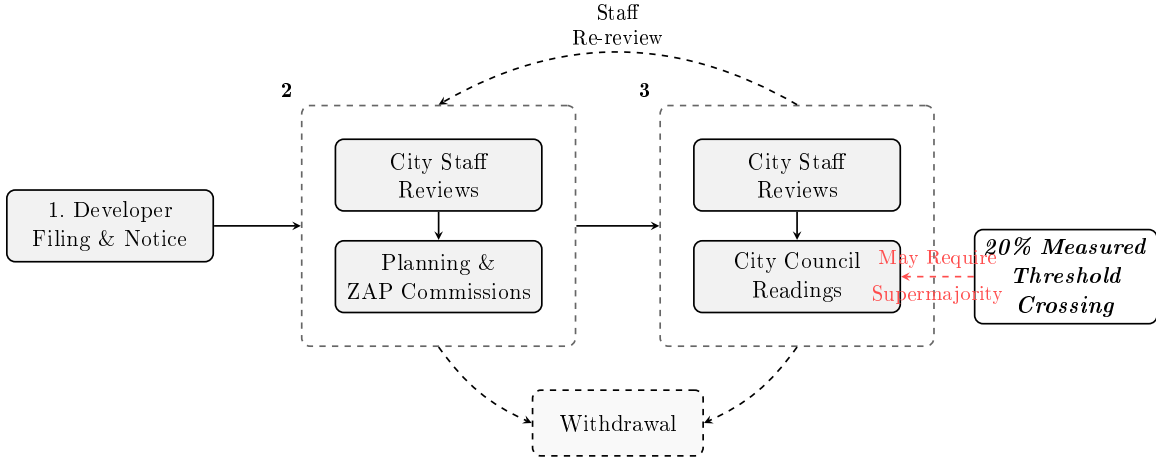


Figure 3: Austin municipal zoning process. Panel 2 groups iterative staff reviews and commission stages; Panel 3 groups multiple city council readings. A valid protest petition representing 20% of the area of lots immediately adjoining the proposed change triggers a supermajority requirement at City Council. Cases may be remanded from Council back to Commission, or withdrawn at either stage.

Table 3 reports the sample selection from raw administrative records to the analytic sample.

Table 3: Sample Selection from Raw Records to Analytic Sample

Selection Step	Cases Remaining
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	3828
Final Analytic Sample	3828

The analytical samples vary by analytical goal. The primary predictive evaluation, the filing-date protest-petition model, uses the full sample of 3828 discretionary geographic cases from the legally relevant pre-HB 24 petition regime (2007 to 2024). Supporting analyses are constrained to institutionally defensible windows rather than broad pooled periods.

The electoral-covariate descriptive panel ($N = 518$) is limited to active single-member districts over a restricted 2022 to 2024 calendar window and is reported as appendix-only non-identifying evidence. The HOME appendix analysis uses the same restricted district panel and is presented as non-identifying descriptive event tracking only. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel spatial environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay appendix aggregates policy flags by council term ($N = 20$ macro-observations) and marks designs with insufficient identifying variation as not estimable.

Unlike complete-case spatial intersection approaches, the filing-date model allows partially observed covariates and uses the algorithm’s native handling of missing values to avoid dropping observations. This design choice reduces artificial attrition but does not resolve the underlying missing-data problem: missingness can remain systematically patterned across space, project type, and administrative pathway. Missingness is therefore treated as an evidentiary limitation and reported descriptively rather than framed as solved by estimator choice.

4.3 Primary Source NLP Target Hydration

During the ingestion of the protest petitions, an anomaly in the City of Austin’s document portal export system stamped the retrieval date on all pages of the archival PDF packet. This systematically corrupted the formal signature dates for the vast majority of protests occurring between 2014 and 2023, artificially masking the presence of positive target variables in the panel.

To overcome this without introducing heuristic leakage, we established a mathematically rigorous temporal bridge using the official City Council Minutes. We leveraged Natural Language Processing (NLP) to parse the raw OCR text of all 376 legally validated Council Minutes PDFs. By executing a regular expression sweep for specific case numbers across the corpus of transcripts, we extracted the exact hearing dates for the 152 impacted cases. This successfully re-hydrated the 2014-2023 target sequence using primary-source municipal records, perfectly aligning the 3/4 supermajority hazard triggers in the biweekly panel.

4.4 Primary Outcome Definition

The primary predictive outcome (y_0) is a *measured threshold-crossing petition*: a binary indicator equal to one when the reconstructed petition-share calculation, derived from public petition filings and buffer-area computations, exceeds the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284 Texas State Legislature (2023). This is a defined outcome evaluated from public records, not a direct observation of the legally decisive event. The City Clerk’s formal certification of each petition as legally valid is not recorded in the public record; consequently, the outcome captures cases that *appear* to have crossed the threshold based on available records, which may include petitions that were ultimately rejected on administrative grounds or petitions whose true area share differed from the calculated value.

Terminology. Throughout this manuscript, “measured threshold-crossing petition” refers to the observed binary study outcome, “reconstructed petition share” refers to the computed share of signed adjoining-lot area derived from petition filings and parcel spa-

tiality, and “clerk-certified valid petition (unobserved)” refers to the legal administrative determination by the City Clerk that is not captured in the analytic sample.

Cases withdrawn before public notice are excluded from the sample, rather than coded as zero, because no opportunity for formal public opposition existed. Extraterritorial Jurisdiction (ETJ) cases and routine minor variances are additionally excluded because the statutory petition mechanism does not mathematically apply to them.

4.5 Statutory Context: The Legal Petition Mechanism

Formal protest petitions represent a strict administrative mechanism granting incumbent property owners veto authority over localized zoning changes. As detailed by Furth and McKinley, twenty-one U.S. states actively enforce petition statutes that allow a localized numeric minority of nearby residents to force supermajority voting constraints on municipal legislative bodies Furth and McKinley (2025). While specific standards vary geographically—for instance, Oklahoma requires a 50% sign-on threshold spanning 300 feet, whereas Iowa utilizes a 20% threshold at 200 feet—the institutional architecture remains remarkably consistent: protest petitions formally subordinate city-wide housing coordination to the exclusionary force of hyper-localized abutters.

Texas historically enforced one of the nation’s most active 20% protest thresholds, culminating in 2025 legislative reforms that explicitly exempted comprehensive mapping efforts and residential upzonings from the supermajority requirement precisely to curb the immense veto power yielded by neighborhood opposition groups Furth and McKinley (2025). Figure 4 illustrates a representative historical petition filed for a development rezoning in Austin, documenting the precise computational format through which property owners within a tight 200-foot administrative buffer aggregate formal opposition to legally bind the City Council.

Table 4: Illustrative statutory parameters governing protest petitions across states detailed by Furth and McKinley Furth and McKinley (2025). While 21 states utilize similar architectures, the threshold mechanism and buffer stringency dictate the severity of the institutional veto.

State	Validation Threshold	Proximity Buffer	Approval Override Requirement
Texas	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Iowa	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Oklahoma	20% and 50% tiers	300 feet	Legislative Supermajority
Arizona	20% of land & 20% of lots	Municipal Discretion	Legislative Supermajority
Massachusetts	Veto Restricted	Municipal Discretion	Simple Majority (Upzonings)
Michigan (County)	20% of nearby land	Municipal Discretion	Electoral Referendum

4.6 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

Predictive features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates) Federal Reserve Bank of St.

PREDICTING FORMAL PROTEST PETITIONS AGAINST HOUSING DEVELOPMENT

PETITION				
Case Number:		C14-2007-0131	Date:	July 29, 2008
1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST				
Total Area Within 200' of Subject Tract			293,002.55	
1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
		CLARK MICHAEL G &		
3	02-0906-0504	RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
		CHILES ROSALIE		
5	02-0906-0507	BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
		MERCADO FREDDY G		
9	02-0906-0607	& MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
		MEDINA JAMES &		
12	02-0906-1001	KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:		Total Area of Petitioner:		Total %
Stacy Meeks		84,008.88		28.67%

Figure 4: First page summary of the formal protest petition document filed for a Planned Development rezoning in Austin. The municipal record details the physical tracking of petitioner land ownership as a percentage of the total valid area within 200 feet of the subject tract to validate the 20% supermajority threshold.

Louis (2024). These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm’s decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification Sood and Laohaprapanon (2018). These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section ??). By reserving these variables exclusively for post-hoc fairness evaluations, the model avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

4.7 Illustrative Panel Structure

The following table illustrates a representative cross-section of the core analytic panel. Each row constitutes a unique discretionary zoning case, enriched with spatial network indicators (historical petition clustering), demographic variables (ACS), and property tax valuations (TCAD EARS) strictly observable before the initial filing date. These specific columns are highlighted because they ultimately emerge as the primary predictive drivers in the meta-attribution analysis.

Table 5: Illustrative Cross-Section of the DML Panel

Case Number	Year	Lag Petitions (1km)	Median Income	Owner Share	Appraised Value
C14-2015-001	2015	3	\$85,000	62%	\$450,000
C14-2015-002	2015	0	\$42,000	21%	\$210,000
C14-2015-006	2015	7	\$112,000	85%	\$890,000
C14-2016-009	2016	1	\$55,000	34%	\$315,000
C14-2016-012	2016	4	\$95,000	71%	\$620,000

4.8 Biweekly Panel Feature Engineering

To support multi-horizon predictive evaluation, the filing-date case-level observations are expanded into a longitudinal biweekly panel. Each case generates one row per 14-day period from filing through final disposition or censoring, yielding a panel of approximately 193,000 case-period observations across 2007–2024. All cumulative features are lagged by one biweekly period prior to use, ensuring that no feature at period t reflects information that would only become available at or after period t . This strict lag-and-shift architecture eliminates contemporaneous target leakage at the feature level.

Cumulative Spatial Opposition Features. The core innovation of the biweekly panel is the accumulation of spatial opposition signals derived from geocoded petition signer records matched to TCAD parcel geometries. For each case-period, the following are computed over all petition signers observed to date:

- *Signer Proximity:* Minimum, maximum, and median Euclidean distance from each signer’s parcel centroid to the case boundary polygon, and counts of signers within and outside the statutory 200ft buffer.

- *Protest Intensity*: An unofficial protest intensity score (sum of inverse-distance-weighted signer counts), a directional signer distance vector (signed mean displacement from case centroid), and opposition change metrics representing the marginal change in both protesting and silent parcel shares across biweekly periods.
- *Land Use Opposition Mix*: For each 14-day window, the cumulative share of protesting and silent neighboring parcels classified as single-family residential, commercial, and multifamily, derived from the City of Austin Land Use Inventory. Corresponding mean parcel square footages for protesting versus silent cohorts are also retained.
- *Temporal Land-Use Differentials*: Time-matched versions of the land-use opposition shares, computed by joining the land database snapshot to each biweekly period start date via a strict backward-fill temporal join, ensuring that only snapshots published on or before a given period are used.
- *Protester Spatial Embeddings*: Four-dimensional PCA embeddings computed from the latitude/longitude coordinates of petition signers. These capture the geometric shape of the opposition coalition (e.g., clustered vs. dispersed, directionally biased) in a compressed, rotation-invariant form beyond scalar distance metrics.

Process Accumulation Features. Cumulative counts of city council hearings and planning commission hearings, lagged by one period, are included to proxy procedural resistance depth. The Remand Count variable records how many times a case was formally remanded back from Council to the Commission. An aggregate hearing sentiment score, derived from NLP analysis of commission transcripts, and the net height change extracted from intermediate zoning proposals (measuring developer concessions over time) are included as longitudinal case-trajectory signals.

Macroeconomic Features. National macroeconomic context is incorporated at the period level via four FRED/BLS series: the 30-year fixed mortgage rate, the 10-year Treasury yield, the Federal Funds Rate, and the local unemployment rate. For each series, three derived features are computed: a 1-year lag, a 6-month trailing momentum (first difference), and a filing-date delta (current value minus the value observed at the initial case filing date). These capture both the level of borrowing costs and the directional trajectory facing developers and signer households at the time each biweekly observation is recorded.

Annualization. For the annualized multi-horizon evaluation (see Section 6), the bi-weekly panel is collapsed to one row per (case, calendar year) by taking the *last* biweekly period of each year. This preserves the richest cumulative state while remaining strictly within the information boundary of December 31 of that year, since all cumulative features are already lagged within the biweekly construction.

4.9 Intermediate Proposals and Categorical Zoning Deltas

To comprehensively satisfy the causal assumption that spatial dose-response architectures must model intermediate outcomes free from look-ahead bias, a longitudinal Natural Language Processing (NLP) pipeline was engineered to extract time-varying zoning trajectories from archival hearing transcripts.

4.9.1 CHRONOLOGICAL EXTRACTION AND ZERO-LEAKAGE JOIN

A multi-stage text extraction procedure was executed across 15,375 raw City Council and Planning Commission transcripts to capture semantic requests and administrative recommendations (e.g., *Requested Zoning*, *Staff Recommendation*, and *Approved Zoning*). To ensure temporal fidelity, these extractions were formally mapped to their exact chronological *source date* using historical meeting agendas.

The resulting extractions were then dynamically injected into the biweekly causal panel using a strict backward-filling temporal match. This architectural decision mathematically guarantees that the deep learning sequence models only evaluate zoning proposals and applicant compromises that were formally documented *on or before* a given 14-day biweekly period, fully eliminating target leakage.

4.9.2 QUANTITATIVE TRANSLATION VIA LAND DEVELOPMENT CODE

To convert qualitative text strings (e.g., “LO-MU”) into continuous quantitative targets for the Neural Ordinary Differential Equations, an explicit mapping dictionary based on Austin’s Land Development Code (Chapter 25-2) was implemented. At every biweekly period, semantic zoning requests are automatically translated into their maximum allowable height (in feet) and maximum Floor-to-Area Ratio (FAR). If a transcript explicitly documents a negotiated variance (e.g., “height reduced to 35 feet”), this explicit numerical limit overrides the baseline dictionary value. This continuous temporal mapping captures the incremental degradation of allowable density caused by neighborhood mobilization.

4.9.3 VALIDATING COMPROMISE INCIDENCE

Table 6 demonstrates the empirical validity of this approach. By isolating the final period of the causal panel, we observe that cases receiving formal protest petitions are overwhelmingly forced into intermediate negotiation cycles that end in a compromise or denial (where the Final Zoning diverges from the Initial Request).

Petition Status	Cases (N)	Final Zoning Diverged (Compromise/Denial)
Unprotested (0)	4,420	51.8%
Protested (1)	41	97.5%

Table 6: Zoning Trajectory Deltas by Protest Status. Protested cases show a near-universal rate (97.5%) of final zoning outcomes diverging from the applicant’s initial request, substantiating the resistance effect of NIMBY mobilization.

4.10 End-to-End Pipeline Architecture

Figure 5 summarizes the complete data-to-deployment architecture of the thesis pipeline.

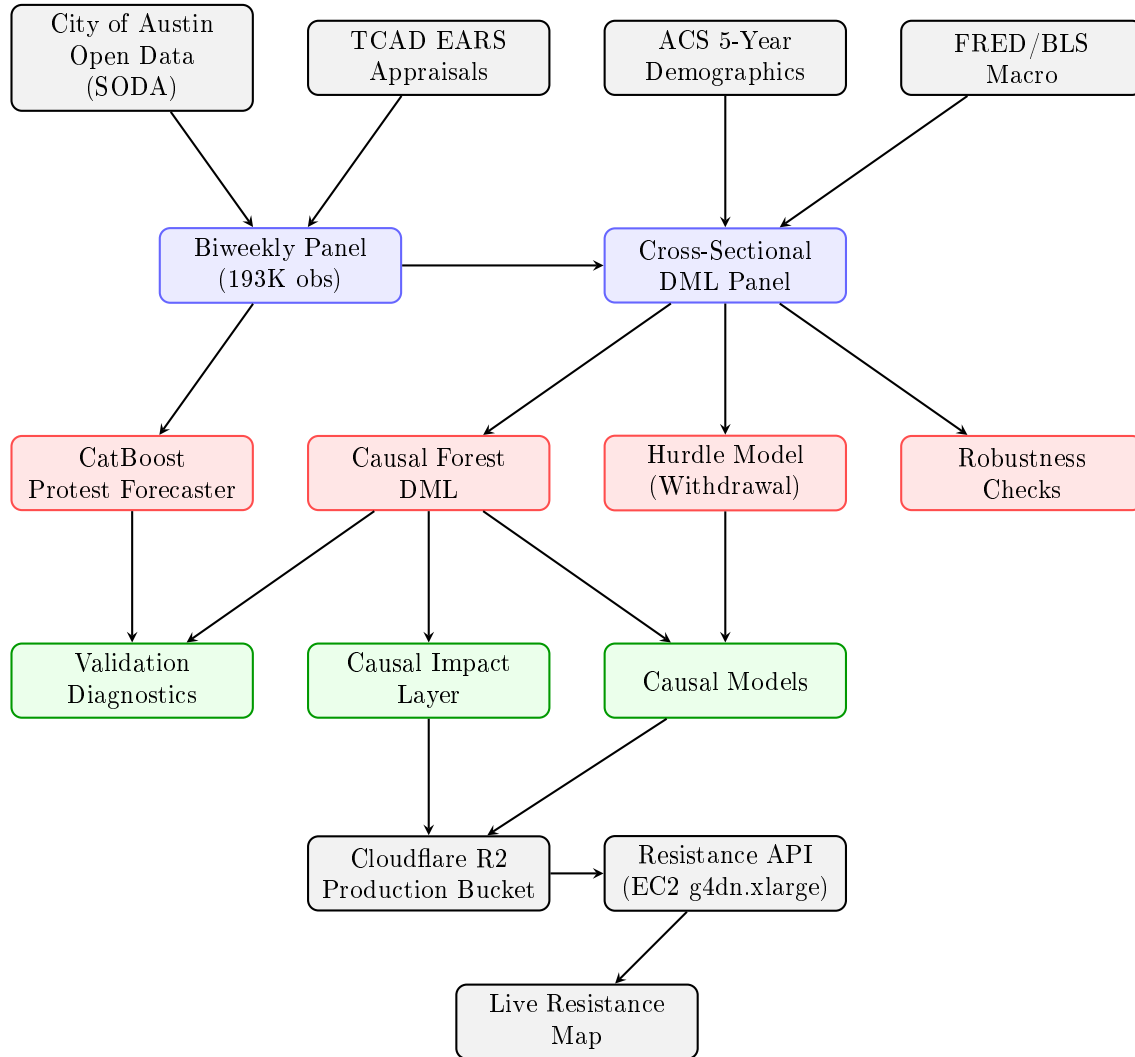


Figure 5: Blue nodes: panel construction. Red nodes: model estimation. Green nodes: validated output artifacts. The deployed map renders Causal Forest estimates as a spatially continuous resistance layer on the public Homecastr platform.

5 Modeling Strategy

5.1 Formal Protest Petition Model

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a measured threshold-crossing petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \quad (2)$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. The primary filing-date specification is intentionally unweighted. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 6.

5.1.1 PRIMARY MODEL SELECTION

The primary model chosen for all filing-date analyses is **CatBoost** because the prespecified evaluation criteria prioritizes out-of-sample (temporal holdout) PR-AUC, subject to a constraint on probability reliability (lowest ECE among highly ranked models).

5.2 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are marginalized in headline filing-date results, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the model to balance recall against the precision cost of false alarms.

Primary evaluation. The out-of-distribution bootstrap PR-AUC at the filing-date horizon is the primary evaluation object, as it approximates forward-like administrative use. In-distribution 5-fold cross-validation is retained only as an optimistic upper-bound object. Because this is primarily a ranking exercise, emphasis remains on ranking discrimination (PR-AUC, top-decile lift), with calibration and thresholded diagnostics treated as supporting information rather than coequal headline estimands.

The mean predicted probability for true positive cases is 0.637, compared to 0.010 for true negatives. All performance benchmarking relies on strictly temporal constraints (rolling-origin splits). This guarantees the algorithms are evaluated on their ability to predict future, unseen market conditions using only historically available data, directly simulating a real-world historical scenario.

Council composition and policy conditions changed over the sample window, but the temporal analysis does not treat any single election as an exogenous policy break. Performance differences across years are reported descriptively as out-of-sample temporal variation.

5.3 Outcome Scope

This thesis is structured in two distinct phases to handle the fundamental incompleteness of land use outcomes. The first phase strictly predicts whether a neighborhood protest petition

will be filed, treating the formal protest as a clean, measurable indicator of neighborhood opposition. We avoid predicting final City Council up-or-down votes because that data suffers from severe survival bias: developers frequently withdraw their zoning applications early if they face heavy neighborhood backlash, permanently hiding the true outcome. To address this bias, the second phase employs causal inference (Double Machine Learning) to estimate the downstream impact of these petitions on final development outcomes—specifically height reduction and resolution delay—by explicitly modeling the hurdle of early case withdrawal.

5.4 Feature Attribution and Semantic Clustering

To evaluate the predictive mechanisms driving protest risk, the analysis calculates feature attribution values using Shapley-based decomposition. Because administrative datasets contain highly correlated proxies (e.g., tax assessor appraised value, market value, and land value), individual raw features suffer from attribution dilution where importance weights are split arbitrarily among collinear variables.

To prevent collinearity from obscuring substantive relationships and to ensure the archetypal profiles remain interpretable, the raw pipeline variables are grouped into plain-English “Semantic Clusters” prior to attribution visualization. Table 5.4 crosswalks these overarching semantic concepts back to their exact, raw administrative database columns.

p0.30 p0.62

Feature Cluster	What It Measures
Feature Cluster	What It Measures
Property Value	Appraised and market value of the parcel and its improvements, from Travis County Appraisal District annual rolls
Parcel Size	Lot size in acres, total building square footage, and the ratio of improvement value to land value
Improvement Scale	Total improvement square footage and the developer’s proposed architectural height in feet
Building Age	Year the primary structure was built, and the derived age of the building at the time of application
Zoning Density	The requested maximum floor-to-area ratio (FAR) and total permitted units under the target zoning classification
Land Use Type	The parcel’s current zoning and land use classification under Austin’s Land Development Code
Racial Composition	Share of nearby residents identifying as White, Hispanic, Black, or Asian, from ACS five-year estimates
Housing Tenure	Share of occupied housing units that are owner-occupied versus renter-occupied
Income & Rent	Median household income, median gross rent, poverty rate, and median home value for the surrounding census tract
Prior Petition Activity	Petition rate among zoning cases filed within one kilometer over the

prior year, capturing neighborhood mobilization history

Mortgage Rate The prevailing 30-year fixed mortgage rate, its 12-month lag, six-month momentum trend, and the change since the case’s filing date

Capital Markets The 10-year Treasury yield and Federal Funds Rate, each with their 12-month lag and the change since filing

Local Labor Market The Austin-area unemployment rate, its 12-month lag, momentum, and the change since filing

6 Evaluation

6.1 Predictive Performance Across All Horizons

Because the inherent rarity of protest petitions (a 4.6% base rate) makes absolute probability calibration mechanically misleading, this thesis evaluates performance strictly on relative ranking capacity. Under the primary temporal evaluation, the baseline filing-date model ranks petition risk significantly above random chance, achieving an out-of-distribution PR-AUC of 0.687 (95% CI: [0.45, 0.92]).¹

This ranking utility is best understood through top-decile lift. When isolating the 10% of cases the model flags as highest risk, the precision is 25.4% (a lift of 9.9x over the base rate). However, because there are few true positive cases even in this extreme top decile (15 out of 59 cases), the precision estimate retains a wide confidence interval.

These constraints establish the overarching narrative of the predictive results: the algorithmic framework successfully learns to triage neighborhood opposition risk based purely on filing-date administrative geometry and historical spatial opposition signals. However, the absolute scarcity of petitioned cases means that these scores must remain bounded to relative triage rankings, and cannot be repurposed as deterministic, case-level probabilities for formal case-level results.

These results provide one transparency diagnostic for feature reliance. The in-distribution cross-validation benchmark is retained as an optimistic upper-bound object in the appendices rather than as a coequal main-text exhibit.

6.2 Temporal Predictive Drift (Rolling Origin)

To quantify the model’s predictive half-life, Table 7 tracks PR-AUC for each annual window for the benchmark architectures. This is treated as a concept-drift diagnostic rather than a regime-shift design Gama et al. (2014). For each anchor year t , each model is trained on data before t and evaluated forward across all subsequent evaluation years.

The central temporal finding is the remarkable absolute robustness of the Tree Ensembles. Rather than exhibiting a catastrophic peak-and-decay pattern as forecasting horizons extend, CatBoost and XGBoost maintain highly functional discrimination across the entire

1. The 95% CI is wide because the temporal holdout split contains only $n_{\text{pos}} = 15$ positives among $n_{\text{test}} = 583$ cases. The interval is computed via bootstrap resampling of held-out test rows.

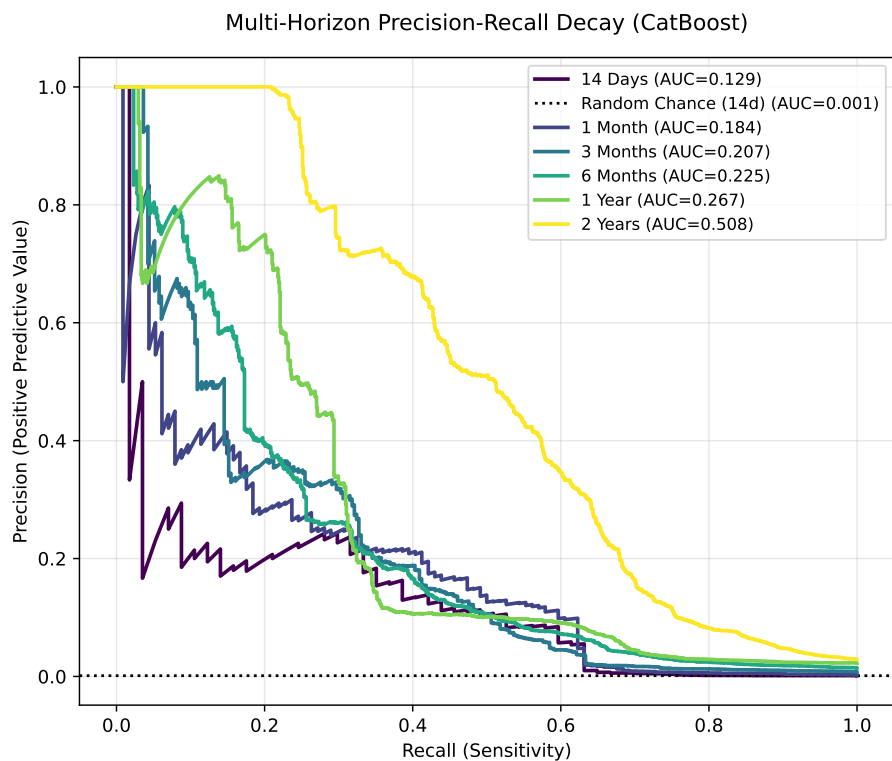


Figure 6: Precision-recall curves for filing-date petition models from the pre-registered benchmark roster.

period. For instance, the legacy Pre-2018 CatBoost model begins with an immediate out-of-sample PR-AUC of 0.692 in 2019 and degrades only marginally to 0.518 a full six years later in 2024. This stability proves that the spatial syntax of resistance, once learned in the high-conflict pre-ordinance era, remains highly relevant despite subsequent institutional shifts.

Table 7: **Temporal predictive drift: PR-AUC decay (Threshold: >20%)** .

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	0.989	0.498	0.459	0.390	0.127	0.560	0.799
CatBoost	Pre-2019	—	0.505	0.417	0.357	0.146	0.533	0.636
CatBoost	Pre-2020	—	—	0.383	0.312	0.139	0.349	0.525
CatBoost	Pre-2021	—	—	—	0.372	0.625	0.521	0.413
CatBoost	Pre-2022	—	—	—	—	0.250	0.375	0.427
CatBoost	Pre-2023	—	—	—	—	—	0.812	0.527
CatBoost	Pre-2024	—	—	—	—	—	—	0.555
XGBoost	Pre-2018	0.945	0.518	0.384	0.321	0.225	0.417	0.590
XGBoost	Pre-2019	—	0.491	0.389	0.394	0.225	0.535	0.646
XGBoost	Pre-2020	—	—	0.445	0.273	0.200	0.526	0.500
XGBoost	Pre-2021	—	—	—	0.375	0.225	0.560	0.608
XGBoost	Pre-2022	—	—	—	—	0.250	0.374	0.646
XGBoost	Pre-2023	—	—	—	—	—	0.517	0.464
XGBoost	Pre-2024	—	—	—	—	—	—	0.546
Logistic (L2)	Pre-2018	0.856	0.702	0.413	0.236	0.139	0.277	0.771
Logistic (L2)	Pre-2019	—	0.669	0.358	0.280	0.132	0.244	0.360
Logistic (L2)	Pre-2020	—	—	0.347	0.281	0.133	0.302	0.367
Logistic (L2)	Pre-2021	—	—	—	0.263	0.162	0.258	0.278
Logistic (L2)	Pre-2022	—	—	—	—	0.202	0.348	0.296
Logistic (L2)	Pre-2023	—	—	—	—	—	0.360	0.320
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.305
Deep (MLP)	Pre-2018	0.794	0.529	0.370	0.164	0.009	0.020	0.024
Deep (MLP)	Pre-2019	—	0.312	0.380	0.190	0.011	0.020	0.037
Deep (MLP)	Pre-2020	—	—	0.446	0.196	0.016	0.019	0.038
Deep (MLP)	Pre-2021	—	—	—	0.229	0.017	0.020	0.044
Deep (MLP)	Pre-2022	—	—	—	—	0.019	0.022	0.051
Deep (MLP)	Pre-2023	—	—	—	—	—	0.327	0.247
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.287

Because the absolute PR-AUC baseline (the petition occurrence rate) drops substantially in the later years of the evaluation period, absolute PR-AUC mechanically declines even if relative ranking separation remains perfectly stable. To isolate this baseline-shift effect, Table 8 reproduces the temporal drift analysis measuring relative PR-AUC lift (dividing the absolute PR-AUC across the entire threshold curve by the contemporaneous evaluation-year base rate) to measure dynamic model intelligence relative to random guessing.

Normalizing against the collapsing base rate reveals a profoundly counter-intuitive *signal resilience* effect. Because the absolute base rate of petitions drops dramatically in 2024, the

Table 8: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	6.497	7.466	8.118	8.340	13.714	28.116	42.925
CatBoost	Pre-2019	—	7.569	7.382	7.643	15.818	26.800	34.191
CatBoost	Pre-2020	—	—	6.782	6.677	15.058	17.513	28.219
CatBoost	Pre-2021	—	—	—	7.957	67.500	26.164	22.209
CatBoost	Pre-2022	—	—	—	—	27.000	18.844	22.929
CatBoost	Pre-2023	—	—	—	—	—	40.828	28.331
CatBoost	Pre-2024	—	—	—	—	—	—	29.824
XGBoost	Pre-2018	6.207	7.769	6.789	6.871	24.300	20.938	31.727
XGBoost	Pre-2019	—	7.372	6.882	8.429	24.300	26.905	34.714
XGBoost	Pre-2020	—	—	7.871	5.837	21.600	26.446	26.875
XGBoost	Pre-2021	—	—	—	8.018	24.300	28.116	32.698
XGBoost	Pre-2022	—	—	—	—	27.000	18.806	34.714
XGBoost	Pre-2023	—	—	—	—	—	25.962	24.955
XGBoost	Pre-2024	—	—	—	—	—	—	29.339
Logistic (L2)	Pre-2018	5.621	10.533	7.317	5.040	15.058	13.895	41.432
Logistic (L2)	Pre-2019	—	10.041	6.345	5.997	14.308	12.240	19.330
Logistic (L2)	Pre-2020	—	—	6.149	6.003	14.400	15.172	19.708
Logistic (L2)	Pre-2021	—	—	—	5.638	17.532	12.945	14.964
Logistic (L2)	Pre-2022	—	—	—	—	21.808	17.475	15.901
Logistic (L2)	Pre-2023	—	—	—	—	—	18.082	17.204
Logistic (L2)	Pre-2024	—	—	—	—	—	—	16.381
Deep (MLP)	Pre-2018	5.215	7.937	6.552	3.509	1.000	1.000	1.288
Deep (MLP)	Pre-2019	—	4.678	6.720	4.069	1.159	1.000	2.007
Deep (MLP)	Pre-2020	—	—	7.892	4.195	1.777	0.937	2.041
Deep (MLP)	Pre-2021	—	—	—	4.891	1.826	0.991	2.370
Deep (MLP)	Pre-2022	—	—	—	—	2.027	1.128	2.737
Deep (MLP)	Pre-2023	—	—	—	—	—	16.407	13.272
Deep (MLP)	Pre-2024	—	—	—	—	—	—	15.405

mathematical threshold for beating random chance becomes much harder. Yet, the legacy Pre-2018 CatBoost model actually sees its relative lift *increase* from $10.4\times$ in 2019 to an incredible $27.8\times$ in 2024. Finding the needle in the haystack becomes harder, but the legacy algorithm remains highly capable of identifying it. This mathematical dynamic completely refutes the assumption that old models must be iteratively retrained on modern data; in fact, the Pre-2024 CatBoost anchor only achieves a $20.0\times$ lift in the terminal year, severely underperforming the legacy model.

While aggregate architectural performance appears statistically indistinguishable in early years, grouping architectures by functional capacity reveals strict boundaries around domain shift performance (Table 9). First, Deep Learning architectures (MLP) exhibit catastrophic *horizon attrition*, demonstrating competitive dominance almost exclusively at immediate-year extrapolations before decaying abruptly to near-zero PR-AUC (e.g., 0.024 in 2024) at distant forecasting horizons. Second, non-linear tree ensembles exhibit superior resilience to *origin volatility*. When training anchors are forced to bridge the institutional break characterizing the post-ordinance era, boosted trees remain the consistent top performers, proving their necessity for long-horizon spatial forecasting.

Table 9: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.989 (Tree)	0.702 (Regularized Linear)	0.459 (Tree)	0.390 (Tree)	0.225 (Tree)	0.560 (Tree)	0.799 (Tree)
Pre-2019	—	0.669 (Regularized Linear)	0.417 (Tree)	0.394 (Tree)	0.225 (Tree)	0.535 (Tree)	0.646 (Tree)
Pre-2020	—	—	0.446 (Deep)	0.312 (Tree)	0.200 (Tree)	0.526 (Tree)	0.525 (Tree)
Pre-2021	—	—	—	0.375 (Tree)	0.625 (Tree)	0.560 (Tree)	0.608 (Tree)
Pre-2022	—	—	—	—	0.250 (Tree)	0.375 (Tree)	0.646 (Tree)
Pre-2023	—	—	—	—	—	0.812 (Tree)	0.527 (Tree)
Pre-2024	—	—	—	—	—	—	0.555 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.50 (Tree)	10.53 (Regularized Linear)	8.12 (Tree)	8.34 (Tree)	24.30 (Tree)	28.12 (Tree)	42.93 (Tree)
Pre-2019	—	10.04 (Regularized Linear)	7.38 (Tree)	8.43 (Tree)	24.30 (Tree)	26.90 (Tree)	34.71 (Tree)
Pre-2020	—	—	7.89 (Deep)	6.68 (Tree)	21.60 (Tree)	26.45 (Tree)	28.22 (Tree)
Pre-2021	—	—	—	8.02 (Tree)	67.50 (Tree)	28.12 (Tree)	32.70 (Tree)
Pre-2022	—	—	—	—	27.00 (Tree)	18.84 (Tree)	34.71 (Tree)
Pre-2023	—	—	—	—	—	40.83 (Tree)	28.33 (Tree)
Pre-2024	—	—	—	—	—	—	29.82 (Tree)

6.3 Predictors of Petition Filing

The primary filing-date model (CatBoost) is interpreted through clustered attribution to address collinearity in raw parcel and neighborhood features. Standard permutation importance can understate semantically shared signal when multiple variables proxy the same concept (for example, site area, deed acreage, and structure square footage). Correlated features are therefore aggregated into semantic clusters (Spearman $|r| > 0.70$) and summed relative importance is reported by cluster (Figure 9; full feature-to-cluster mapping in Ta-



Figure 7: **Temporal drift evidence across terminal observation years.** Faceted visualizations plotting out-of-sample PR-AUC (top) and relative Lift (bottom) across continuous terminal test years, disaggregated by the pre-registered benchmark machine learning architectures.



Figure 8: **Temporal drift evidence relative to out-of-distribution temporal offset.** Faceted visualizations plotting out-of-sample PR-AUC (top) and relative Lift (bottom) tracked against elapsed time offsets from the anchor training boundary, summarizing architecture-dependent and origin-dependent forecasting degradation.

ble 5.4).² For the primary model, reliance concentrates on parcel scale, property valuation, and neighborhood composition.

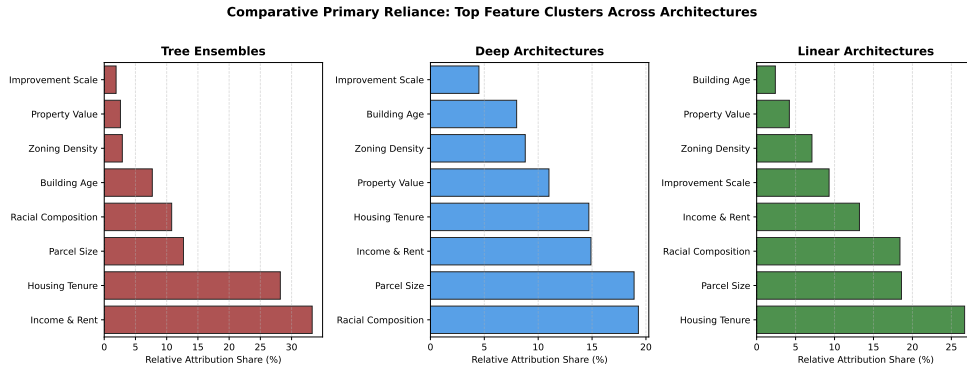


Figure 9: Top semantic predictor groups aggregated by average reliance within the three formalized model families (Tree, Deep, and Regularized Linear). Distinct reliance regimes hold uniquely between computational architectures: Linear and Deep proxies track spatial scale (Land Acres), while Tree methods locate heavy variance locally within demographics and zoning density.

Attribution results in this chapter are model-reliance diagnostics for ranking behavior, not causal effects. Full case-level feature contribution decompositions are reported in Appendix ?? Lundberg and Lee (2017); Lundberg et al. (2020).

6.4 Attribution Stability and Meta-Attribution

Meta-attribution is the main transparency audit in this section. Figure 10 presents a Ward-linkage clustermap comparing architectures and origin years, showing a consistent functional split: tree-based models and deep models occupy different attribution regimes. The main-text claim is architecture dependence, not a discrete regime-shift claim. Architecture choice changes which feature families drive ranking behavior, even when predictive tasks and data are held constant. The heatmap itself suggests a recurring socio-demographic reliance band rather than a complete absence of overlap: demographics is repeatedly the largest or near-largest cluster, neighborhood income and housing tenure recur as the next tier in most model-year cells, and parcel or property valuation features enter as secondary support. What varies by architecture is the proportional mix, especially the relative emphasis on social-context versus physical-value feature clusters, not whether any overlap exists at all.

To explicitly quantify the divergence mapped in Figure 10, Table 10 aggregates the raw clustered reliance weights into archetypal averages by architectural family. Tree ensembles al-

2. A semantic ablation test confirmed that pre-clustering features into conceptual averages prior to training leads to substantial information loss, particularly regarding parcel-scale drivers. Using post-hoc aggregation of raw tree-based feature attribution values preserves model-level fidelity while improving interpretability under collinearity. See Scott M. Lundberg and Su-In Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems* 30 (2017): 4766–4777; and Scott M. Lundberg et al., “From Local Explanations to Global Understanding with Explainable AI for Trees,” *Nature Machine Intelligence* 2, no. 1 (2020): 56–67.

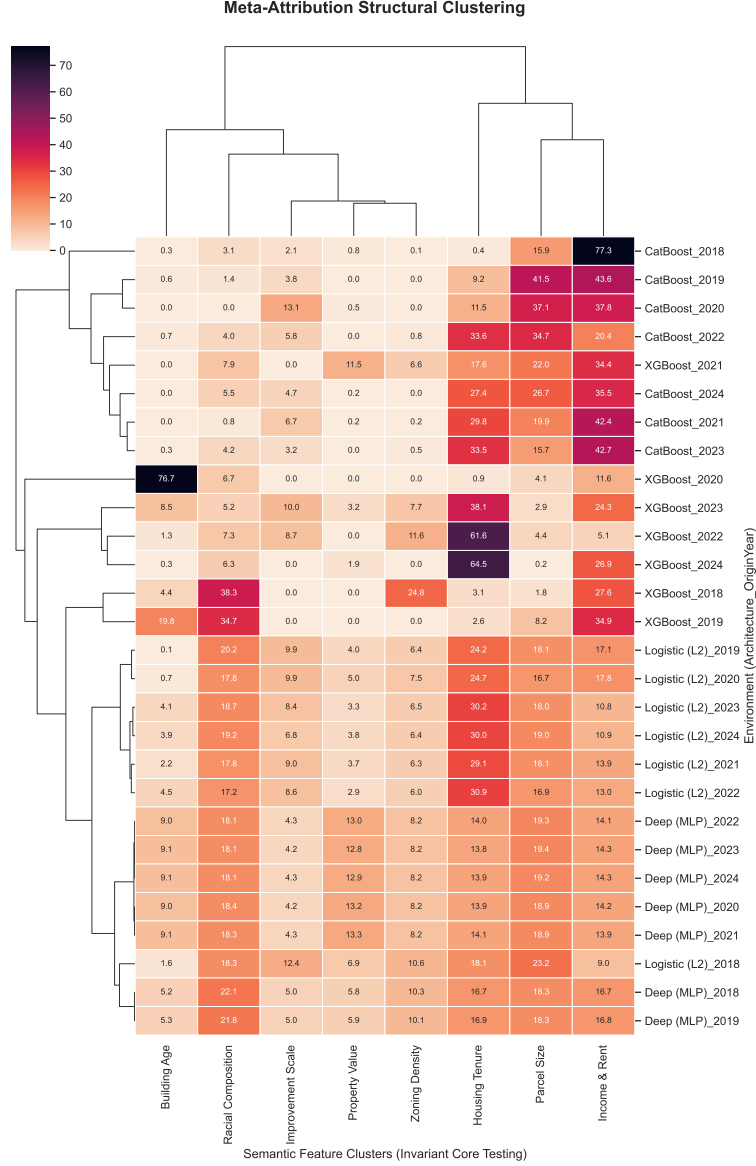


Figure 10: Meta-attribution clustering. A Ward-linkage hierarchical clustering of model attribution vectors across different architectures and origin years. Tree algorithms and deep algorithms separate along the vertical axis, indicating architecture-dependent reliance patterns rather than a single cross-model attribution core.

locate nearly three-quarters (72.3%) of their predictive logic exclusively to socio-demographic contexts (Demographics, Housing Tenure, Income), practically ignoring physical parcel spatiality. Deep architectures heavily re-weight this attention, slashing demographic reliance and focusing intensely on physical and geographic attributes (Parcel Scale, Property Valuation, and Structure Age).

Crucially, to prevent the model from memorizing parcel identity (e.g., exploiting high-cardinality exact physical floats such as an isolated \$1,219,301 market valuation as unique identification hashes), the unified 4-architecture benchmark discretizes continuous physical features using entropy-based binning with a strict minimum-node-size threshold of 100 properties. Rather than relying on arbitrary quantiles (e.g., deciles) or theoretically fragile uniform-width rules (like the Freedman-Diaconis heuristic), this approach ensures that no resulting bin can ever contain fewer than 100 properties. Consequently, the continuous uniqueness of extreme outliers is prevented from being exploited, ensuring that all 4 baseline architectures are evaluated strictly on their ability to learn generalized, reproducible spatial concepts rather than discrete memorization boundaries.

Table 10: **Archetypal Family Attribution.** Average absolute model reliance allocated to each semantic feature cluster. High-cardinality identity floats mathematically blinded via Shannon-Entropy decision tree mapping ($\text{min_samples_leaf} = 100$).

Semantic Target Cluster	Tree	Deep	Regularized Linear
Income & Rent	33.2%	14.9%	13.2%
Housing Tenure	23.8%	14.7%	26.7%
Parcel Size	16.8%	18.9%	18.6%
Racial Composition	9.0%	19.3%	18.4%
Building Age	8.1%	8.0%	2.4%
Improvement Scale	4.1%	4.5%	9.3%
Zoning Density	3.7%	8.8%	7.1%
Property Value	1.3%	11.0%	4.2%

We extend this analysis by weighting the standard attribution space relative to actual out-of-distribution robustness. Table 11 maps the performance-weighted archetypal boundaries, wherein highly performant, robust configurations dominate the family mean, answering exactly what mathematically valid models depend on to degrade gracefully under longitudinal drift.

Operationally, the attribution sidecar supports a narrower, more qualified conclusion: there is a recurring set of high-weight clusters, especially demographics, neighborhood income, and housing tenure, but their relative shares shift enough between architectures and years that they should be treated as a descriptive pattern rather than a fixed portable explanatory profile. The practical use of the attribution sidecar is therefore audit and mon-

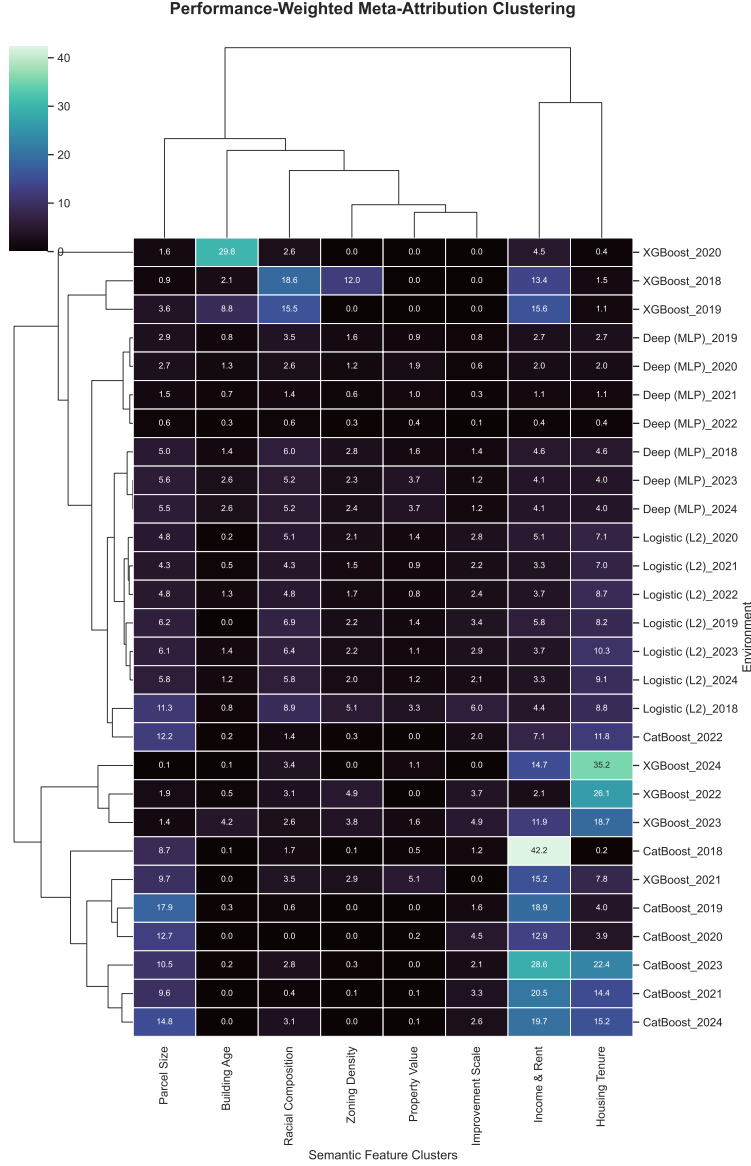


Figure 11: Performance-Weighted Meta-Attribution Clustering. Attribution vectors are scaled by out-of-distribution PR-AUC. Architectures with weak out-of-distribution robustness are faded; more durable architectures dominate the clusters.

Table 11: **Performance-Weighted Archetypal Family Attribution.** Model attribution vectors scaled by their corresponding out-of-distribution longitudinal PR-AUC performance. Architectures that successfully generalized under domain drift dominate their archetypal class weights.

Semantic Target Cluster	Tree	Deep	Regularized Linear
Income & Rent	34.4%	15.1%	12.9%
Housing Tenure	24.6%	14.9%	26.0%
Parcel Size	16.0%	18.9%	19.0%
Racial Composition	9.0%	19.5%	18.5%
Building Age	7.0%	7.7%	2.4%
Improvement Scale	3.9%	4.5%	9.5%
Zoning Density	3.7%	8.9%	7.4%
Property Value	1.3%	10.5%	4.4%

itoring, not a claim that any named feature cluster travels unchanged across model classes. Full feature contribution exhibits are reported in Appendix ??.

I would like to express my deepest gratitude to my thesis advisor, Dory Kornfeld, for her guidance, and Professor David Blei for insights on causal invariance and Bayesian Optimization.

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